

Homework 5: Cluster Analysis

S&DS 5360 | Multivariate Statistics

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Setup & Data Overview

We continue our work with the Healthy Minds Survey (HMS) data. The variables are continuous, but lie on different scales due to the construction of each variable (a sum of underlying Likert-scale questions). Our total number of observations is $n = 68,062$.

- Flourishing (`flourish`): 8–56
- Depression (`deprawsc`): 0–27
- Anxiety (`anx_score`): 0–21
- Loneliness (`lonesc`): 3–9
- Disordered Eating (`ed_sde`): 0–5

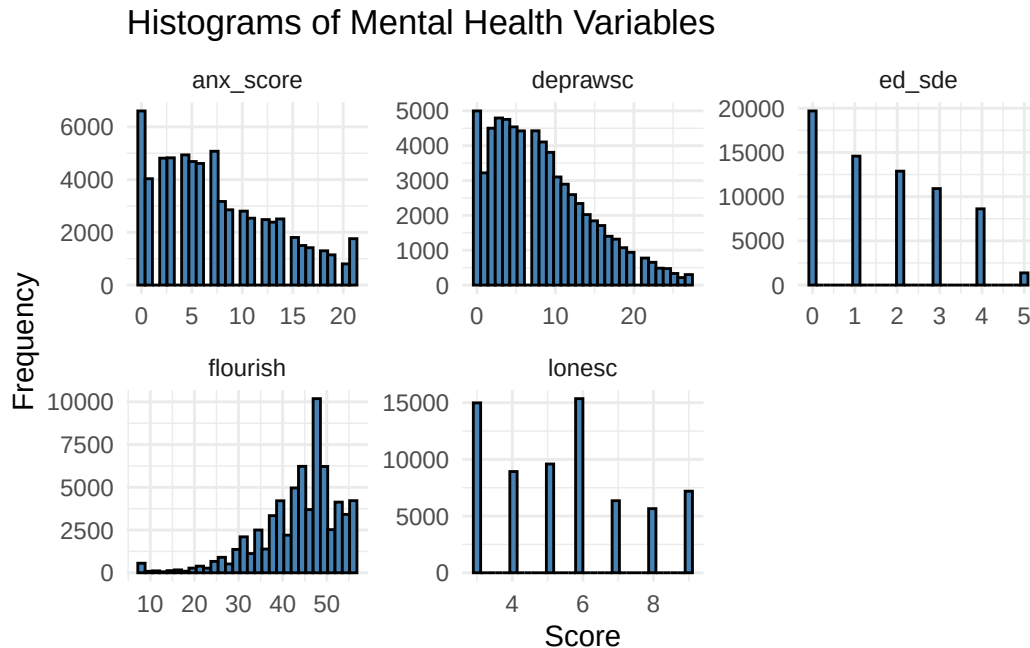
We omit the code to preprocess the data.

Part 1 | Distance Metrics, Standardization, and Transformation

Because we are clustering strictly on continuous, numeric mental health summary scores, standard geometric distance metrics are the most appropriate for our data. As such, we will elect to use the traditional Euclidean distance (straight-line distance between students' standardized score profiles). As a secondary metric, we elect to use Manhattan distance, which is less sensitive to extreme outliers.

First, recalling our previous analyses on this dataset, we may need to transform our variables to normalize our data.

```
df %>%
  pivot_longer(cols = everything(), names_to = "Variable", values_to = "Score") %>%
  ggplot(aes(x = Score)) +
  geom_histogram(bins = 30, fill = "steelblue", color = "black") +
  facet_wrap(~ Variable, scales = "free") +
  theme_minimal() +
  labs(title = "Histograms of Mental Health Variables",
       x = "Score", y = "Frequency")
```



The histograms for anxiety, depression, and disordered eating all exhibit right skewness. Flourish exhibits a left skew, and loneliness retains the underlying ordinal structure from its smaller subset of questions. To correct these, we propose a log transformation to the right-skewed variables and a square transformation to the left-skewed variable. We present the resulting Q-Q plots following the transformations:

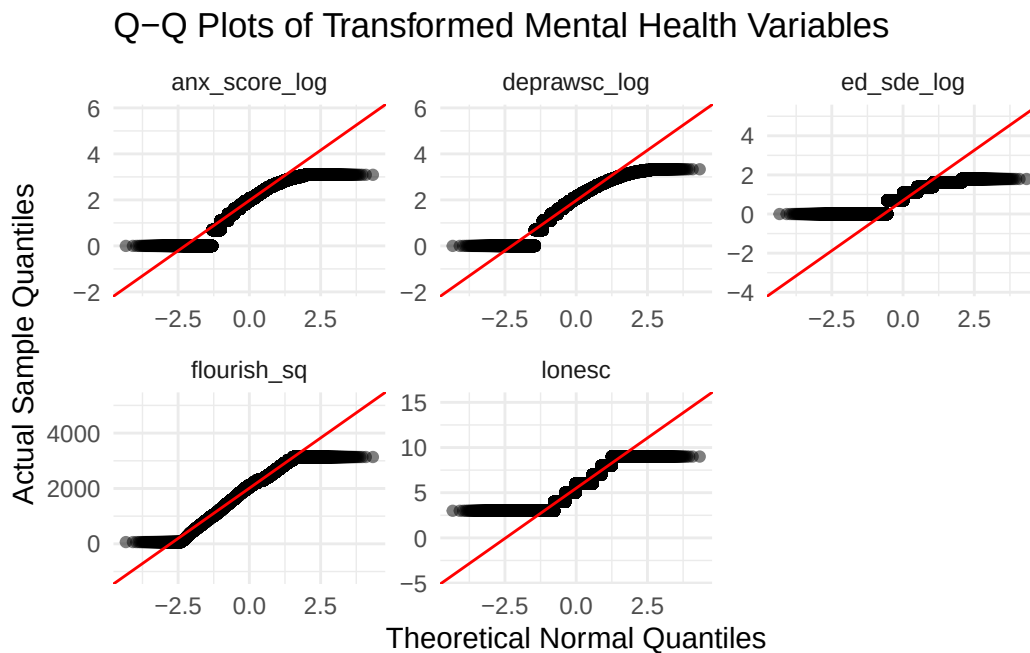
```
df_transformed <- df %>%
  mutate(
    deprawsc_log = log(deprawsc + 1),
    anx_score_log = log(anx_score + 1),
    ed_sde_log = log(ed_sde + 1),
    flourish_sq = flourish^2) %>%
  select(-deprawsc, -anx_score, -ed_sde, -flourish)
```

```

par(mfrow = c(2,3))

df_transformed %>%
  pivot_longer(cols = everything(), names_to = "Variable", values_to = "Score") %>%
  ggplot(aes(sample = Score)) +
  stat_qq(color = "black", alpha = 0.5) +
  stat_qq_line(color = "red") +
  facet_wrap(~ Variable, scales = "free") +
  theme_minimal() +
  labs(title = "Q-Q Plots of Transformed Mental Health Variables",
       x = "Theoretical Normal Quantiles",
       y = "Actual Sample Quantiles")

```



We see a staircase effect in the Q-Q plots, which is inherent to the bounded survey data we are working with; transformations cannot change this. However, a bulk of our observations follow the Q-Q line, and observe that our post-log transformed variables exhibit lighter tails, meaning our outliers were pulled inward. This ensures that extreme outliers will not disproportionately inflate the Euclidean distance calculations between students. We see a similar effect with our squared `flourish` variable.

As can be observed in the overview in the previous section, each continuous variable exists along a different scale. We will need to standardize our variables; else, flourishing and other

variables with large ranges will dominate. To standardize, we convert our scores to Z-scores ($Z \sim N(0,1)$).

```
df_scaled <- scale(df_transformed)
```

Part 2 | Hierarchical Cluster Analysis

With our data transformed and scaled, we now perform hierarchical cluster analysis. Given we are working with continuous data, we will compare both Euclidean and Manhattan distances across Ward's, complete, and average linkage:

- **Euclidean + Ward's:** The literature supports this as the most balanced approach for our continuous data. We will use this as our baseline for comparison with the other methods.
- **Euclidean + Complete Linkage:** Controlling for our distance metric, this comparison allows us to isolate how changing the agglomeration method (from minimized variance to furthest neighbor) changes our clusters.
- **Manhattan + Complete Linkage:** Controlling for agglomeration method, we can see how changing our distance metric impacts the tree.
- **Manhattan + Average Linkage:** Another comparison to further analyze the outlier-resistant nature of Manhattan distance.

Note that we will need to sample our data to $n = 5,000$ for computational feasibility. Our true sample size exceeds the maximum limitation for the `hclust()` function.

```
# Sample n = 5,000
set.seed(4747)
df_sample <- as.data.frame(df_scaled) %>% sample_n(5000)

# Calculate Distance Metrics
dist_eucl <- dist(df_sample, method = "euclidean")
dist_manh <- dist(df_sample, method = "manhattan")

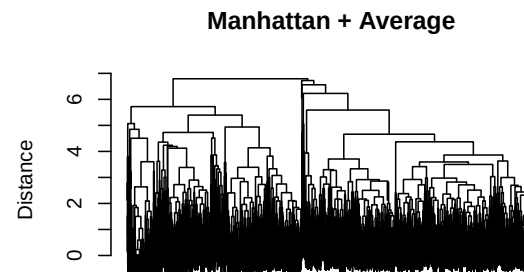
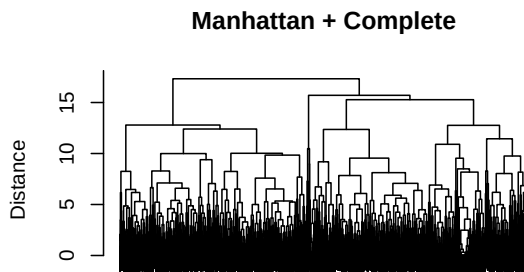
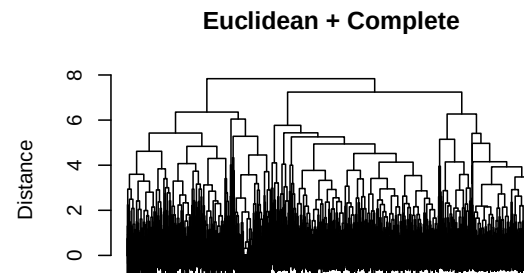
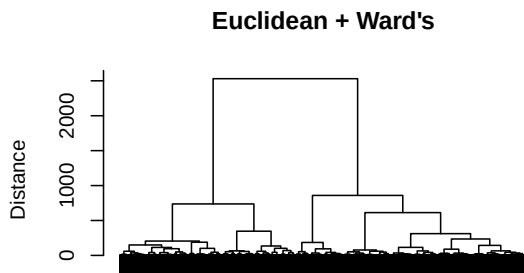
# Run Cluster Analysis
hc_eucl_ward <- hclust(dist_eucl, method = "ward.D")
hc_eucl_comp <- hclust(dist_eucl, method = "complete")
hc_manh_comp <- hclust(dist_manh, method = "complete")
hc_manh_avg <- hclust(dist_manh, method = "average")

# Plot
par(mfrow = c(2, 2))
```

```

plot(hc_eucl_ward, labels = FALSE, cex = 0.5,
     xlab = "", ylab = "Distance", main = "Euclidean + Ward's", sub = "")
plot(hc_eucl_comp, labels = FALSE, cex = 0.5,
     xlab = "", ylab = "Distance", main = "Euclidean + Complete", sub = "")
plot(hc_manh_comp, labels = FALSE, cex = 0.5,
     xlab = "", ylab = "Distance", main = "Manhattan + Complete", sub = "")
plot(hc_manh_avg, labels = FALSE, cex = 0.5,
     xlab = "", ylab = "Distance", main = "Manhattan + Average", sub = "")

```



The resulting dendrograms visibly demonstrate how different agglomeration methods handle dense survey data like ours. Euclidean + Ward suggests between 3–5 distinct clusters. When we switch to complete linkage, instead of forming distinct tree trunks, the data clumps into a massive, dense blob, with small individual branches very close to the top. Generally, complete and average linkage methods were difficult to interpret, suffering from severe chaining and highly unbalanced tree structures with poorly defined separation at higher distances. Meanwhile, Euclidean distance paired with Ward's method produced a highly interpretable den-

derogram, minimizing within-cluster variance and separating the data into distinct, balanced branches.

Part 3 | Discussion on Groups

To confirm the 3–5 clusters that we visually assessed from the Euclidean + Ward analysis in Part 2, we mathematically evaluate the number of clusters needed via `NbClust` and the courses's custom `HClusEval`. We perform this on a small sample of data ($n = 1,000$) for computational purposes.

```
set.seed(4747)
df_sample <- as.data.frame(df_scaled) %>% sample_n(1000)

par(mfrow = c(1, 2))

eval <- hclus_eval(df_sample, dist_m = 'euclidean', clus_m = 'ward',
  plot_op = T, print_num = 10)
```

```
[1] "Creating Distance Matrix using euclidean"
```

```
[1] "Clustering using ward"
```

```
[1] "Clustering Complete. Access the Cluster object in first element of output"
```

```
[1] "Calculating RMSSTD"
```

```
[1] "RMSSTD Done. Access in Element 2"
```

```
[1] "Calculating RSQ"
```

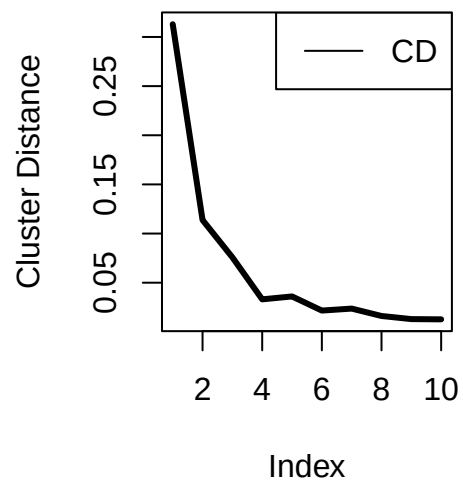
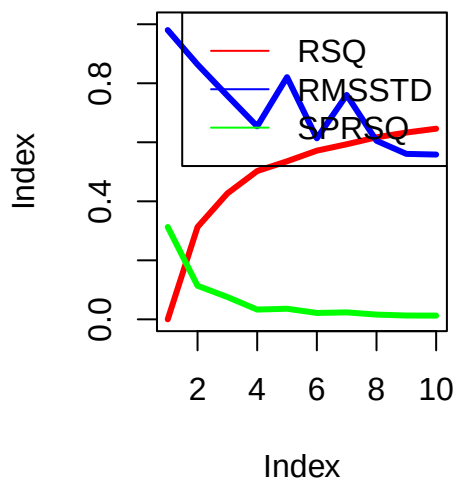
```
[1] "RSQ Done. Access in Element 3"
```

```
[1] "Calculating SPRSQ"
```

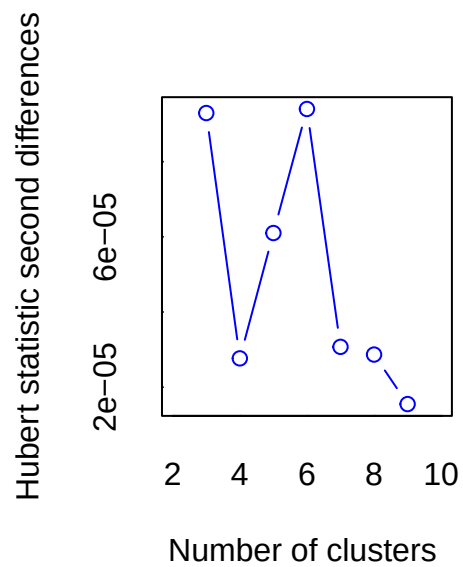
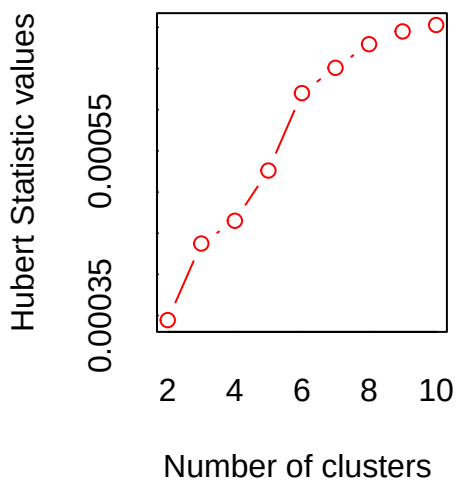
```
[1] "SPRSQ Done. Access in Element 4"
```

```
[1] "Calculating Cluster Dist. "
```

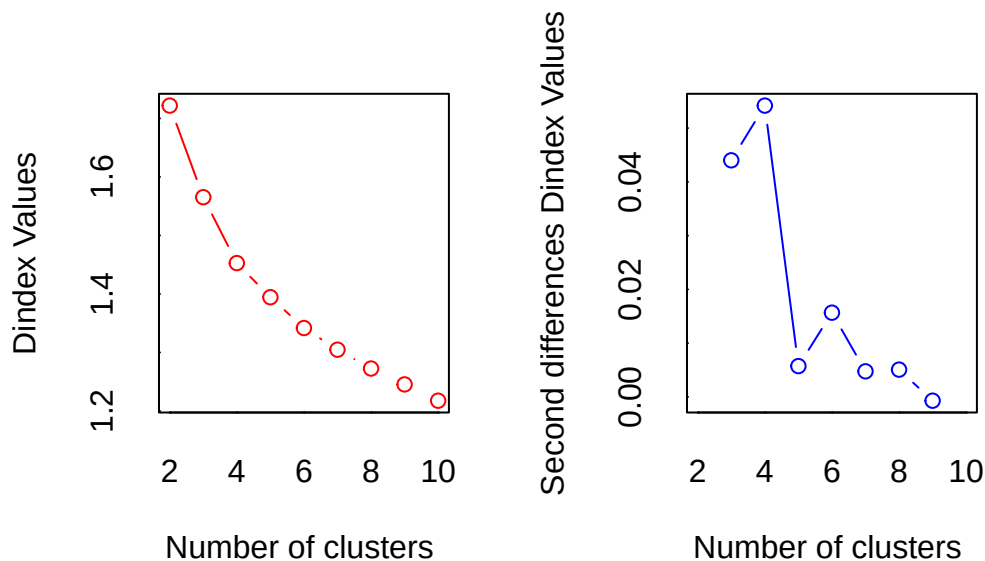
```
[1] "CD Done. Access in Element 5"
```



```
nb_results <- NbClust(df_sample, distance = 'euclidean', min.nc = 2,
                      max.nc = 10, method = 'ward.D', index = 'all',
                      alphaBeale = 0.1)
```



*** : The Hubert index is a graphical method of determining the number of clusters.
 In the plot of Hubert index, we seek a significant knee that corresponds to a significant increase of the value of the measure i.e the significant peak in index second differences plot.



*** : The D index is a graphical method of determining the number of clusters.
 In the plot of D index, we seek a significant knee (the significant peak in second differences plot) that corresponds to a significant increase of the value of the measure.

* Among all indices:
 * 7 proposed 2 as the best number of clusters
 * 2 proposed 3 as the best number of clusters
 * 11 proposed 4 as the best number of clusters
 * 1 proposed 7 as the best number of clusters
 * 1 proposed 8 as the best number of clusters
 * 1 proposed 10 as the best number of clusters

***** Conclusion *****

* According to the majority rule, the best number of clusters is 4

Visual inspection of the evaluation plots reveal a distinct elbow in the R- Squared (RSQ) trajectory at $k = 4$ clusters, with diminishing returns in explained variance beyond this point. The Hubert statistic and D-index second difference plots also exhibited significant peaks at $k = 4$. The NbClust majority rule confirms our visual assessment, concluding that $k = 4$ is the optimal number of clusters based on 11 voting indices. For reference, a visualization of these four clusters is plotted below:

```
par(mfrow = c(1, 1))
cuts <- cutree(hc_eucl_ward, k = 4)
plot(hc_eucl_ward, labels = FALSE, cex = 0.6, xlab = "", ylab = "Distance",
      main = "Clustering of HMS Students", sub = '')
rect.hclust(hc_eucl_ward, k = 4)
```

Clustering of HMS Students



We proceed with $k = 4$ for our final k -means partitioning.

Part 4 | k -Means

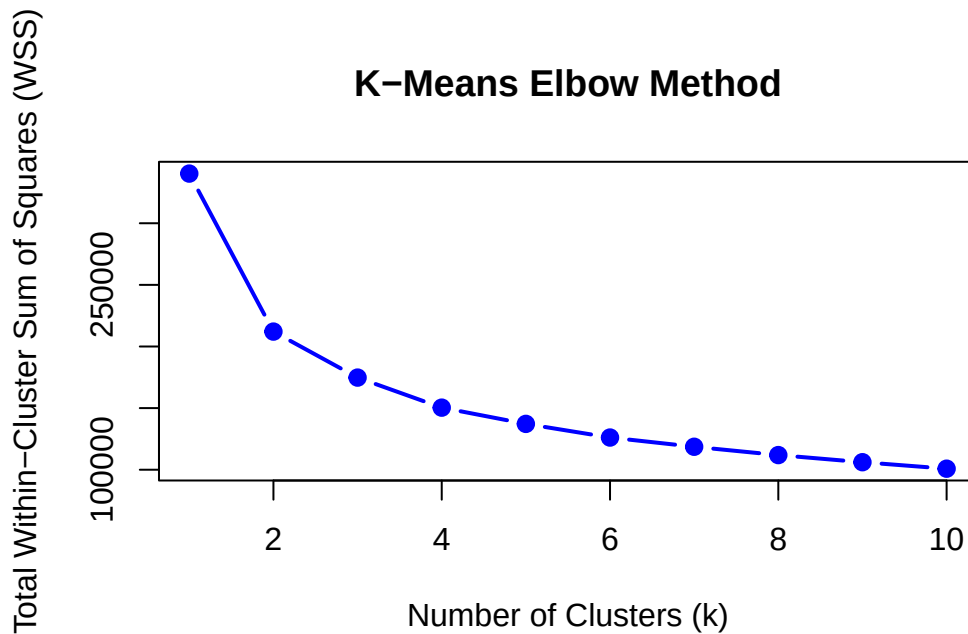
We perform k -means clustering on our data and compare with our results from Part 3. Note that to ensure model convergence with our large sample size, we use the MacQueen algorithm with 50 iterations.

```
set.seed(4747)

wss <- numeric(10)

for (k in 1:10) {
  wss[k] <- kmeans(df_scaled, centers = k, nstart = 25,
                  iter.max = 50, algorithm = "MacQueen")$tot.withinss
}

plot(1:10, wss, type = "b", pch = 19, col = "blue", lwd = 2,
     xlab = "Number of Clusters (k)",
     ylab = "Total Within-Cluster Sum of Squares (WSS)",
     main = "K-Means Elbow Method")
```



```
final_kmeans <- kmeans(df_scaled, centers = 4, nstart = 25,
                      iter.max = 50, algorithm = "MacQueen")
```

```
df_final <- df %>%
  mutate(Cluster = as.factor(final_kmeans$cluster))
```

Our Total Within-Cluster Sum of Squares (WSS) plot exhibits a distinct elbow at $k = 4$, thereafter which the steep reduction in variance begins to plateau. When comparing these results to the hierarchical analysis in Step 3, the findings are consistent (though an elbow at $k = 2$ could also be discerned in this method). Thus, both the objective mathematical indices and the visual k -means elbow plot independently suggest $k = 4$ natural groupings within our data.

Steps 5–6 | Visualization in PCA Space

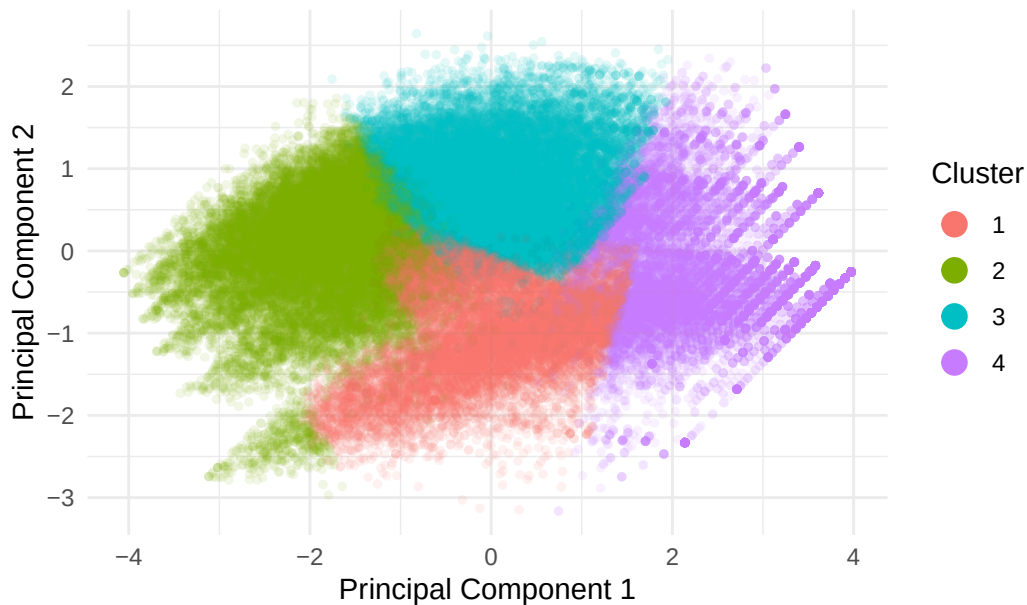
Because the data we are working with exists in 5-dimensional space, we collapse our data in to just two dimensions to help us visualize our groupings. While we could use both PCA and discriminant analysis, both methods of which we've previously used on this dataset, we elect to use PCA because it is an unsupervised method, whereas discriminant analysis is supervised. That is, PCA will serve as an unbiased canvas for us to visualize our groups on.

```
pca <- prcomp(df_scaled)

pca_data <- data.frame(
  PC1 = pca$x[, 1],
  PC2 = pca$x[, 2],
  Cluster = as.factor(final_kmeans$cluster)
)

ggplot(pca_data, aes(x = PC1, y = PC2, color = Cluster)) +
  geom_point(alpha = 0.1, size = 1) +
  guides(color = guide_legend(override.aes = list(alpha = 1, size = 4))) +
  theme_minimal() +
  labs(title = "K-Means Clusters Visualized via PCA (k=4)",
       x = "Principal Component 1",
       y = "Principal Component 2",
       color = "Cluster")
```

K-Means Clusters Visualized via PCA (k=4)



We see four distinct clusters in along our two-dimensional principal axes, confirming our previous analyses. While some boundary overlap exists, this is expected; the clusters themselves are largely continuous with dense central hotspots. We do observe some striping within the clusters, notably cluster #3 (in blue), but this is likely an artifact of the discrete, bounded nature of the underlying Likert-scaled survey data.

Step 7 | Final Analysis of Groups

With multiple independent analyses confirming that $k = 4$ groups exist within our data, we endeavor to assign a meaning to each group. To do this, we present the average scores per variable, per cluster:

```
cluster_profiles <- df_final %>%
  group_by(Cluster) %>%
  summarise(
    Count = n(),
    Mean_Flourish = round(mean(flourish, na.rm = TRUE), 2),
    Mean_Depress = round(mean(deprawsc, na.rm = TRUE), 2),
    Mean_Anxiety = round(mean(anx_score, na.rm = TRUE), 2),
    Mean_Eating = round(mean(ed_sde, na.rm = TRUE), 2),
    Mean_Lonely = round(mean(lonesc, na.rm = TRUE), 2)
  )
```

Table 1: Final K-Means Cluster Profiles (k=4)

Cluster	Student Count	Flourishing	Depression	Anxiety	Eating Concerns	Loneliness
1	14098	42.37	8.53	7.91	0.27	5.77
2	19016	35.54	14.45	12.53	2.71	7.47
3	20389	47.74	7.02	6.92	2.49	4.76
4	14559	49.71	1.69	1.38	0.59	3.76

Most notably, Cluster 4 stands out as our thriving baseline, exhibiting the highest flourishing score and, conversely, extremely low scores across all negative psychological variables. Meanwhile, Cluster 2 seems to represent our most high-risk population, with the lowest flourishing score and highest scores across negative psychological variables. Cluster 3 is unique: while at first glance, these students appear to be high-functioning, exhibiting high flourishing scores and relatively low depression and anxiety, their eating concern score is dramatically elevated compared to baseline. Thus, they likely form an at-risk group unique to disordered eating. Finally, Cluster 1 seems to be comprised of the opposite: relatively elevated depression, anxiety, and loneliness, but with the lowest eating concern score. This seems to indicate a population dealing with generalized, sub-clinical internalizing symptoms who do not engage in disordered eating behaviors.